

ICE Workbench

What decides the sign of x , why the coefficients belong in the change row, and when the “ x is small” shortcut is allowed.

Name _____

Date _____

Period _____

Demonstration: andresdbp.github.io/resources/demos/ice-workbench/

1 Before you open anything

Answer from what you already know. The point is to have something committed to paper that the demonstration will later settle one way or the other.

PREDICTION

A sealed flask at 425 °C contains 0.100 mol L⁻¹ HI and nothing else — no H₂, no I₂. For $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2 \text{HI}(\text{g})$, $K = 54.3$ at that temperature, which is a large number.

Does x come out positive or negative, and is $[\text{HI}]$ at equilibrium higher or lower than the 0.100 you started with?

Write one sentence saying what you used to decide.

2 Reading one complete ICE table

Open the demonstration. It starts on the **ICE table** tab. The concentration sliders move in steps of 0.002 mol L⁻¹.

1. In **The reaction**, choose $\text{H}_2 + \text{I}_2 \rightleftharpoons 2 \text{HI}$.
2. Press **Back to the reference K**, then confirm the **K** readout.
3. Set $[\text{H}_2]_0 = 0.1000$, $[\text{I}_2]_0 = 0.1000$, $[\text{HI}]_0 = 0.0000$.
4. Copy the four boxes in **The reaction** panel into the first table, then read the sentence under the Q-versus-K line and the **Equilibrium** row of the ICE table for the second.

Reference K	at	ΔH°	Δn_{gas}

Q	Forward or backward?	x (mol L ⁻¹)	Equilibrium [H ₂]	Equilibrium [I ₂]	Equilibrium [HI]

a. In the row labelled **Change – in symbols**, what is written under HI, and why is it not the same as what is written under H₂?

b. Multiply your value of x by 2 and compare it with the equilibrium [HI]. Show the arithmetic, then say in one sentence why those two numbers have to agree.

c. Tick **Show the most common wrong answer**. Record the [HI] on the **Wrong answer** row:
 _____ mol L⁻¹. That answer still satisfies $K = [\text{HI}]^2 / ([\text{H}_2][\text{I}_2])$. State what it does violate.

d. Leave the box ticked and switch the reaction to **CO + H₂O ⇌ CO₂ + H₂**. What happens to the two red rows, and what is it about this reaction that causes that?

3 What actually sets the sign of x

One slider moves; everything else stays where it is. Work down the table in order.

1. Choose **N₂O₄ ⇌ 2 NO₂** and press **Back to the reference K**. The **K** readout should show 0.00460.
2. Leave **[N₂O₄]₀** at **0.1000** for every row.
3. Set **[NO₂]₀** to each value in the left-hand column and record what the demonstration reports.

$[\text{NO}_2]_0$	Q	Q larger or smaller than K?	Runs forward or backward?	x, with its sign	Equilibrium $[\text{NO}_2]$
0.0000					
0.0100					
0.0200					
0.0300					
0.0400					

a. Compute Q by hand for the fourth row from $Q = [\text{NO}_2]^2 / [\text{N}_2\text{O}_4]$. Show the working and compare with the value you recorded.

b. Between which two rows does the sign of x change? Working to the nearest 0.002, find the setting of $[\text{NO}_2]_0$ that makes x smallest in magnitude, and write down what the sentence under the Q-versus-K line says there.

c. K held the same value in every row. Complete the rule: the sign of x is decided by _____, and not by _____.

d. Go back to your part 1 prediction and test it: switch to $\text{H}_2 + \text{I}_2 \rightleftharpoons 2 \text{HI}$ and press **Products only**. Was your prediction right? If your reasoning used the size of K, say what that reasoning leaves out.

4 When the “x is small” shortcut fails

Assuming the consumed concentrations barely move turns a quadratic into a square root. It is a legitimate move, and it comes with a condition.

1. Set up $\text{H}_2 + \text{I}_2 \rightleftharpoons 2 \text{HI}$ again with $[\text{H}_2]_0 = 0.1000$, $[\text{I}_2]_0 = 0.1000$, $[\text{HI}]_0 = 0.0000$ and **Back to the reference K**.
2. Open the **Exact vs shortcut** tab and record the first row below.
3. Press **Move K until it is safe**, record the second row; then press **Move K until it breaks** and record the third.

Setting	K	x exact	x shortcut	Percent error	5% rule check
Reference K					
Moved until safe					
Moved until it breaks					

a. Press **Back to the reference K** again and look at the **Side by side** table. What does the Shortcut column give for $[\text{H}_2]$, and what does the row **Q of that answer** report for that column? Explain in one sentence what has gone wrong physically.

b. Still at the reference K, read the **Multiply it out** panel. What degree is the equation in x, how many roots does it report, and on what grounds is one of them thrown out? Note carefully whether the rejected root is a negative number.

c. With H_2 and I_2 in the flask, does the shortcut improve as K rises or as K falls? Now press **Products only** and answer the same question. Why is there no single rule about the direction?

The K slider is a thought experiment, not a laboratory control. Read the note directly underneath it: the only thing that changes K for a given reaction is temperature, and for some settings the note will tell you that no temperature produces that K at all.

5 Solve these without the demonstration

Close the page or look away from it. Show your working. Then reopen it, set each situation up, and check yourself – and where you disagree, find the step where it happened.

1. $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2 \text{HI}(\text{g})$, $K = 54.3$. A flask starts with 0.200 M H_2 , 0.200 M I_2 and no HI. Find x and all three equilibrium concentrations. (Take a square root rather than expanding the quadratic, and say why you are allowed to.)

2. $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2 \text{NO}_2(\text{g})$, $K = 0.0046$ at 25 °C. A flask starts with 0.100 M N_2O_4 and no NO_2 . (a) Solve for x with the “ x is small” shortcut. (b) Apply the 5% check. (c) If it fails, solve the quadratic properly and report the percent error the shortcut would have carried.

3. For $\text{H}_2 + \text{I}_2 \rightleftharpoons 2 \text{HI}$ with $K = 54.3$, a flask holds $[\text{H}_2] = 0.0500 \text{ M}$, $[\text{I}_2] = 0.0500 \text{ M}$ and $[\text{HI}] = 0.400 \text{ M}$. Without solving for x , compute Q and state the direction of reaction, the sign of x , and whether $[\text{H}_2]$ rises or falls.

4. Starting from 0.100 M H_2 , 0.100 M I_2 and no HI, someone writes the change row as $-x$, $-x$, $+x$, solves, and obtains concentrations that satisfy $K = [\text{HI}]^2 / ([\text{H}_2][\text{I}_2])$ exactly. Explain why the answer must still be rejected, and name the quantity that is not conserved by it.

6 On your own

Nothing here is graded. The **Le Chatelier** tab is untouched by everything above.

- On the **Le Chatelier** tab choose **Add a catalyst** and drag **Catalyst speed-up** from 1 to 25. Watch the **K before** and **K after** readouts and the ends of the solid and dashed curves. Decide exactly what a catalyst does change.
 - Choose **Change volume**. Predict the shift from the **Δn gas** readout before you look, first for $\text{H}_2 + \text{I}_2 \rightleftharpoons 2 \text{HI}$ and then for $\text{N}_2 + 3 \text{H}_2 \rightleftharpoons 2 \text{NH}_3$.
 - Choose **Change temperature** — the one stress that moves K itself. Compare an exothermic reaction with the endothermic $\text{N}_2\text{O}_4 \rightleftharpoons 2 \text{NO}_2$: which way does K go when each is heated, and how does the sign of ΔH° predict it?
 - Choose **Add product** on the water-gas shift. Does the species you added come back down to the value it had before the stress? Say what that means for the phrase “the system opposes the change”.
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